Tristan Visser

FINAL-YEAR BSC COMPUTER SCIENCE STUDENT · BACKEND AND FULL-STACK DEVELOPER

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PROFILE

Final-year Computer Science student at Stellenbosch University with a strong foundation in backend architectures, relational database design, REST APIs and Agile workflows. I've built a graph-based dependency engine with Electrum engineers, a full-stack hackathon platform, a reliable UDP file-transfer protocol, and database-backed web applications for commercial clients. Looking for a graduate software engineering role from 2027.

EDUCATION

2023 — 2026

BSc Computer Science

Stellenbosch University

Focus: backend architectures, relational databases, REST APIs, software engineering.

EXPERIENCE

Freelance

Commercial Web Developer & Technical Consultant

Local business clients

- Designed, developed and deployed custom web applications backed by relational databases and secure hosting.
- Managed requirements, domain and DNS configuration, responsive UI and ongoing maintenance.

2026

Dependency Analysis Framework

Electrum · industry project

Built a Java dependency graph engine that identifies breaking API changes and opens automated GitHub pull requests for affected repositories, across three Agile sprints.

SELECTED PROJECTS

2026

Electrum Dependency Analysis Framework

A full-stack dependency analysis framework that maps Maven graphs, classifies release breakage, and opens version-bump pull requests in dependency order.

JAVA 21 · SPRING BOOT · REACT · REACT FLOW · MAVEN · GITHUB API · GITLAB CI

Case study

2026

Hackathon Hosting Platform

A full-stack hackathon platform with accounts, groups and teams, a collaborative editor, sandboxed Python evaluation and live leaderboards.

REACT 19 · VITE · EXPRESS · PRISMA · SQLITE · WEBSOCKETS · NODE.JS · DOCKER

Case study

2026

P2P File Sharing & RBUDP

A peer-to-peer file-sharing application with hybrid RSA/AES encryption, and a custom reliable UDP protocol benchmarked against TCP.

JAVA · JAVAFX · UDP · TCP · RSA / AES · TAILSCALE

Case study

SKILLS

LANGUAGES	Java · Python · JavaScript · SQL · C · Lua
BACKEND AND DATA	Spring Boot · Node.js · Express · Prisma · PostgreSQL · Supabase · REST APIs · WebSockets
FRONTEND	React.js · Next.js · Tailwind CSS · HTML · CSS · WordPress
SYSTEMS AND TOOLS	C/OpenMP · MPI · Multi-threading · UDP/TCP · JavaFX · Linux/Bash · Git · Docker · Cloudflare · Vercel · JVM
WAYS OF WORKING	Agile / Scrum · Semantic versioning · CI/CD · Requirements gathering · Compiler design

ELSEWHERE

GitHub LinkedIn Email

LAST UPDATED: OCTOBER 2026 · REFERENCES AVAILABLE ON REQUEST.